

Zenish Borad

Backend & Blockchain Engineer

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Summary

Backend and blockchain engineer focused on trading and on-chain systems whose results can be **independently verified**. Implemented **ERC-8004** end to end, including a **Validation Registry** where the official deployments include none, and built Bitcoin full-node RPC pipelines into normalized SQL and Solidity contracts with fork, fuzz, and full-coverage tests.

Technical Skills

Blockchain & Web3: Ethereum EVM Internals, Solidity Smart Contracts, ERC-20 / ERC-721 / ERC-8004, UUPS Proxies & Upgradeability, Uniswap v2/v3 AMMs, Bitcoin Core RPC, keccak256 Commitments & ECDSA Signing

Programming Languages: Python, Solidity, SQL, JavaScript

Backend & Web3 Tools: Foundry (forge, cast, anvil), Hardhat, Web3.py, Ethers.js, REST APIs, Flask, React, LLM Text-to-SQL

Data & Quantitative: Pandas, NumPy, Scikit-learn, Predictive Modeling, Backtesting, Risk Metrics (Sharpe, Drawdown)

Cloud & Infrastructure: AWS, Docker, Git, GitHub Actions (CI), SQLite, PostgreSQL

Education

Northeastern University

Master of Science in Information Systems — GPA: 3.7 / 4.0

Sep 2025 – May 2027

Boston, MA

Parul University

Bachelor of Technology in Computer Science and Engineering

2020 – 2024

Vadodara, India

Projects

ChainProof: Verifiable Trading Agent

Jan 2026 – Sep 2026

Solidity, Foundry, Python, Web3.py, Flask, pytest, GitHub Actions

- Built an on-chain verification layer on **Ethereum Sepolia**: hashed 30 trading decisions over their input snapshots with **keccak256**, signed each with **ECDSA**, and committed them through a Solidity registry; a **standalone verifier** checks every hash and signature (**30/30**) from a clean clone, with no keys or configuration
- Implemented **ERC-8004** (draft standard for trustless agent identity): registered the agent as an **ERC-721** identity, then wrote and deployed a **Validation Registry** and validator, since the official deployments include none; the validator **binds verdicts to on-chain state**, so a result cannot be published for an uncommitted decision hash
- Wrote **19 Foundry tests**, including fuzz tests, against **forked live Sepolia registries**, plus **14 Python tests**, all run in **CI**; surfaced that the canonical registry omits a function the reference implementation calls, so it reverts in production
- Built the Python decision engine (RSI, MACD, SMA crossover, Bollinger Bands; trades when 2 of 4 agree) with stop-loss/take-profit, and a **backtest evaluator** adding fees, a **buy-and-hold benchmark**, and **out-of-sample tests**

Bitcoin Text-to-SQL Analytics Pipeline

Python, Bitcoin Core RPC, SQLite, Gemini API

- Built a pipeline syncing **48,000+ transactions** from a **Bitcoin full node** into a normalized SQLite schema spanning blocks, transactions, inputs, outputs, and addresses, then layered a **natural-language-to-SQL engine** (Gemini API) over it, returning correct SQL on **75% (9/12)** of a self-designed query benchmark

SimpleAMM: Constant-Product AMM & Web3 UI

Solidity, Hardhat, React, Ethers.js

- Implemented a **constant-product AMM** in Solidity with a 0.30% swap fee and transferable **ERC20 LP shares**, backed by a Hardhat test suite at **100% line and branch coverage**
- Deployed a live **React/Ethers.js** interface on Sepolia — deposit, redeem, and swap flows, a live $x \cdot y = k$ reserves curve, and swap price history reconstructed from **on-chain event logs via eth_getLogs**

Experience

Sigzen Technologies Pvt. Ltd.

Jan 2024 – Aug 2024

Backend Engineer Intern

Ahmedabad, India

- Consolidated inventory and sales data from **three separate business systems** into a single normalized dataset using **Python (Pandas, NumPy)**, replacing a manual spreadsheet process repeated every reporting cycle
- Containerized the workflow with **Docker** so the same pipeline ran identically across development and production, making scheduled runs reproducible rather than machine-dependent
- Built **SQL dashboards** over the unified dataset, giving product and finance teams direct query access to trend data they had previously assembled by hand